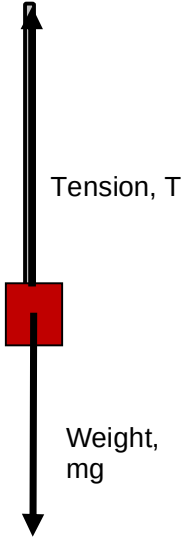
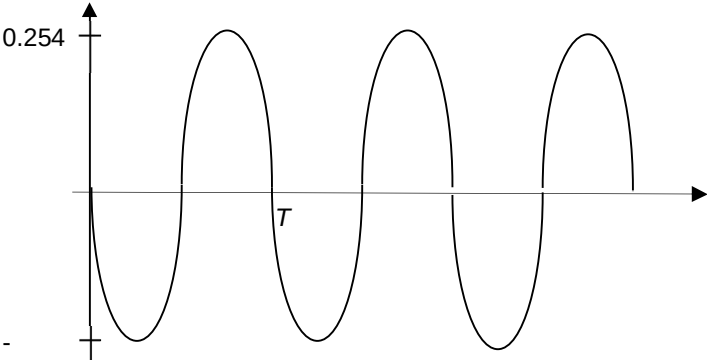


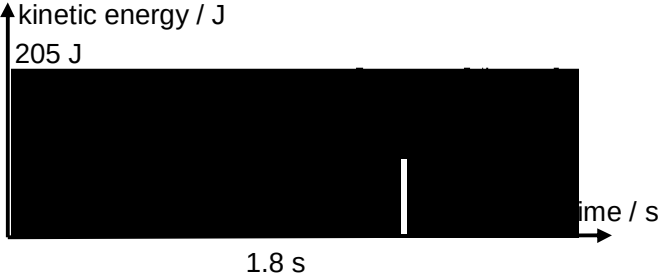
$$= 3.97 \times 10^6 \text{ m s}^{-1}$$

9 (a)

Velocity is the rate of change of displacement of the body.

Angular velocity is the rate of change of angular displacement of the body in oscillations.

9 (b)(i)		
9 (b)(ii)	By conservation of energy, Loss in GPE = Gain in KE $mg(0.6 - 0.6\cos 6^\circ) = \frac{1}{2}mv^2$ $v = 0.254 \text{ m s}^{-1}$	
9 (b)(iii)	$T - mg = \frac{mv^2}{r}$ $T = \frac{mv^2}{r} + mg$ $= \frac{(0.8)(0.254)^2}{0.6} + 0.8(9.81) = 7.93 \text{ N}$ Tension in rod is equal to the force on pivot.	
9 (b)(iv)	velocity, v/ms^{-1} 	

9 (c)(i)	When the driving frequency is equal to the natural frequency of the oscillating system, maximum energy is transferred from the periodic force to the system, which causes the system to oscillate with maximum amplitude . This phenomenon is called resonance .	
9 (c)(i) 1.	$x = x_o \sin \omega t$ $= 0.18 \sin\left(\frac{2\pi}{T}\right)t$ $= 0.18 \sin 3.52t$	
9 (c)(i) 2.	$a = -\omega^2 x $ $= (3.52)^2 (0.08)$ $= 0.99 \text{ m s}^{-2}$	
9 (c)(i) 3.	$KE_{\max} = \frac{1}{2} m \omega^2 x_o^2$ $= \frac{1}{2} (1020) (3.52)^2 (0.18)^2$ $= 205 \text{ J}$	
9 (c)(i) 4.		
10(a)	The half life $t_{1/2}$ of a radioactive nuclide is the average time	